

Answer the following questions based on today's lesson.

Q1. What is enterprise software? Explain in your own words.

Enterprise software is software for large-scale businesses or organizations. It is software that is designed to handle/manage the operations, data and processes of a large-scale business/organization. We can think of it as software built for whole organization rather than one individual use.

Q2. What does a software engineer need to think about besides writing code? Name and briefly explain the important areas discussed in today's lesson.

A software engineer's work isn't just to write code and make the codes work. Following are the things they need to think about besides writing codes:

a. Requirements

They need to be mindful about what problem they are solving, who will use the application, what features are actually necessary, what are the functional requirements and what are the constraints.

b. User experience

Software engineer should make sure that the users can actually use the application easily and make sure the UI is responsive.

c. Architecture

They should decide how the system be structured. They should decide technologies for frontend, backend, authentication and how APIs will communicate.

d. Security

They should make sure the application is safe and users' information is secure and there is safe data handling. They should consider: Authentication, authorization, data privacy, password protection, input validation, SQL/NoSQL injection, XSS(Cross site scripting), secure APIs, secure payments.

e. Scalability

They also need to think about what happens if the system gets bigger. Hence, they need to make sure that the application is scalable by avoiding architectural decisions that will make the system's growth unnecessarily difficult.

f. Performance

g. Testing

h. Maintainability

- i. Costs
- j. Business Requirements
- k. Future changes
- l. Communication and teamwork

Q3. How do different software systems communicate with each other? Explain the role of an API and give one simple example.

Different software systems communicate mainly via APIs (Application programming interface).

Role of an Api

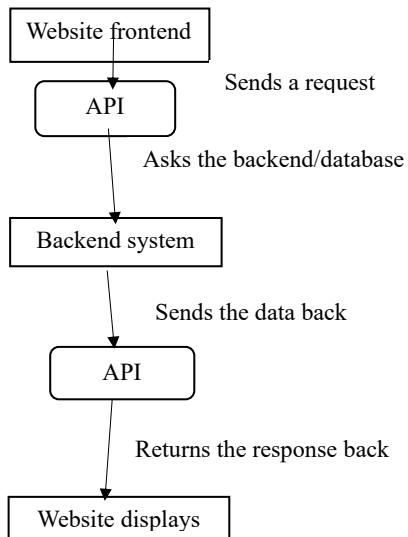
APIs act like a bridge between two software systems. It defines how one system can request data from another system and how the response should be returned.

Common example is REST API

- a. GET: retrieve data
- b. POST: send/create data
- c. PUT/PATCH: update data
- d. DELETE: remove data

EXAMPLE

A simple ecommerce website



Q4. How does software respond to events and protect users? Explain events, authentication, and authorization with simple examples.

Software responds to events by detecting an action or occurrence and then performing an appropriate response.

- Event: An action or occurrence that the software detects.
 - *Example:* A user clicks the Login button. The click is an event, and the software responds by checking the login details.
- Authentication: The process of verifying who a user is.
 - *Example:* A user enters their email and password. The system checks whether the credentials are correct. If they are correct, the user is authenticated.
- Authorization: The process of determining what an authenticated user is allowed to do.
 - *Example:* An employee can view customer information, but only an administrator can delete customer accounts.

Q5. Why do software systems need standards and communication protocols? Explain how they help different systems work together.

Software systems need standards and communication protocols so that different systems can communicate and understand each other correctly.

- Standards are agreed rules or formats that define how information should be represented or exchanged.
- Communication protocols are rules that specify how data is transmitted between systems.

For example, when a web browser accesses a website, HTTP/HTTPS provides rules for how the browser and web server communicate.

Another example is JSON, a commonly used data format for exchanging information between applications.

Without common standards and protocols, different systems might use different formats or communication methods and would have difficulty understanding each other.

ERP Practical Exercise: FreshGrocers, CleanGreen & Odoo

Case Study 1: FreshGrocers

Q1. Identify **six key business requirements** of FreshGrocers that could be addressed by an ERP system.

Six key business requirements of FreshGrocers are:

Inventory management- Track stock levels across the stores and manage perishable goods

Sales and POS- Process customer purchases and payments and record sales transaction and also updates the stock

Customer relationship management (CRM)- Maintain customer information, purchase history

Supplier Management- Manages supplies, purchase orders, deliveries and supplier payments

Financial Management- Track income, expenses, payments and generate financial reports

Human Resources (HR) Management- manages employee records, attendance, payroll, and recruitment.

Q2. For each requirement identified in Question 1, recommend a suitable ERP module and briefly explain its purpose.

Inventory management:

Inventory Management Module- Tracks stock levels across stores, manages reordering, and helps reduce waste, especially for perishable goods.

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Sales and POS:

Sales and POS Module- Processes sales and payments and automatically updates inventory after each transaction.

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Customer Management:

CRM Module- stores customer information, purchase history, preferences, and supports loyalty programs and promotions.

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Supplier Management:

Supplier Module- Manages suppliers, purchase orders, deliveries, and supplier-related payments.

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Financial Management:

Financial Management Module- Records financial transactions, manages expenses and revenue, and generates financial reports.

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Human Resources (HR) Management:

HR Module- Manages employee records, attendance, payroll, and recruitment.

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Q3. A customer purchases several products from a FreshGrocers store. Explain how this transaction could be processed through an integrated ERP system. Identify the relevant ERP modules involved.

The transaction could flow in following ways:

1. The sales/pos module records the products purchased by the customer
2. System calculates the total price and processes the customer's payment
3. The inventory management module automatically reduces the stock quantities for the products that are sold.
4. If stocks falls below a certain level the syste can notify the supplier module to reorder the supplies
5. The CRM module can update the customer's purchase history if customer is registered.
6. The financial management module recordsthe sales revenue and payment accounting purposes
7. Managers can use the erp system to generate sales, inventory, and financial reports

Relevant modules:

Sales/POS → Inventory → CRM → Procurement → Finance

Q4. Explain **two benefits** FreshGrocers could achieve by integrating its business processes through an ERP system.

1. Improves efficiency and reduce manual work;
An ERP system automatically shares information between departments. For example, when a product is sold, the inventory and financial records can be updated automatically. This reduces duplicate data entry and human errors.
2. Helps in better decision-making;
Managers can access up-to-date information about sales, inventory, finances, and customers in one system. This helps them make better decisions about purchasing, stock levels, promotions, and business performance.

Case Study 2: CleanGreen Laundry Services

Q5. Identify **six key business requirements** of CleanGreen Laundry Services that could be addressed by an ERP system.

The six key requirements are:

1. **Order Management** – Track laundry orders from receiving them through cleaning to delivery.
2. **Inventory Management** – Manage detergents, cleaning supplies, and packaging materials.

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3. **Scheduling and Delivery Management** – Schedule customer pickups and deliveries and manage routes.
4. **Customer Relationship Management (CRM)** – Store customer details, preferences, order history, and feedback.
5. **Financial Management** – Manage payments, expenses, payroll, and financial reporting.
6. **Human Resources (HR) Management** – Manage employees, attendance, payroll, and recruitment.

Q6. For each requirement identified in Question 5, recommend a suitable ERP module and briefly explain its purpose.

Business Requirement	ERP Module	Purpose
Order Management	Order Management Module	Tracks orders from receiving the customer's request to cleaning and final delivery.
Inventory Management	Inventory Management Module	Tracks detergents, cleaning supplies, and packaging materials across different locations.
Scheduling & Delivery	Scheduling/Delivery Management Module	Schedules pickups and deliveries and helps organize efficient delivery routes.
Customer Management	CRM Module	Stores customer details, preferences, order history, and feedback.
Financial Management	Financial Management Module	Tracks payments, revenue, expenses, payroll, and generates financial reports.
Human Resources	HR Module	Manages employee information, attendance, payroll, and recruitment.

Q7. A customer places an online laundry order. Describe the main business process from receiving the order to delivering the cleaned clothes to the customer. Identify the ERP modules that could support each stage of the process.

The process could work as follows:

1. Online Order Received

The customer places an order through the online system. The Order Management Module records the customer's details, laundry requirements, and order information.

2. Pickup Scheduled

The Scheduling and Delivery Management Module schedules a pickup and assigns the appropriate delivery staff.

3. Clothes Received

The customer's clothes are collected and the order status is updated in the system.

4. Cleaning Process

The laundry staff processes the clothes. The Order Management Module tracks the order's progress and status.

5. Inventory Used

Detergents, cleaning supplies, and packaging materials are used. The Inventory Management Module updates the stock levels.

6. Payment and Financial Record

The customer's payment is recorded by the Financial Management Module, which also tracks the revenue from the order.

7. Delivery Scheduled

Once cleaning is completed, the Scheduling and Delivery Management Module schedules the delivery.

8. Cleaned Clothes Delivered

The clothes are delivered to the customer and the order status is updated as completed.

Q8. Explain two benefits CleanGreen could achieve by integrating its business processes through an ERP system.

1. Better order tracking and service quality:

The ERP system can track an order throughout the entire process, from pickup to cleaning and delivery. This reduces the risk of lost or delayed orders and helps CleanGreen provide more consistent service.

2. Improved resource and financial management:

The company can monitor cleaning supplies, employee information, payments, expenses, and revenue in one system. This helps reduce shortages, control costs, and make better business decisions.

Comparative Analysis

Q9. Compare FreshGrocers and CleanGreen Laundry Services. Identify **two ERP requirements that are common to both businesses** and **two requirements that are different**.

Two common ERP requirements

1. Inventory Management

Both businesses need to track and manage their supplies.

- **FreshGrocers:** Tracks food and other products available for sale.
- **CleanGreen:** Tracks detergents, cleaning supplies, and packaging materials.

2. Financial Management

Both businesses need to manage their financial transactions.

- **FreshGrocers:** Records sales, payments, expenses, and financial reports.
- **CleanGreen:** Records customer payments, expenses, payroll, revenue, and financial reports.

Two different ERP requirements

1. Sales and POS – FreshGrocers

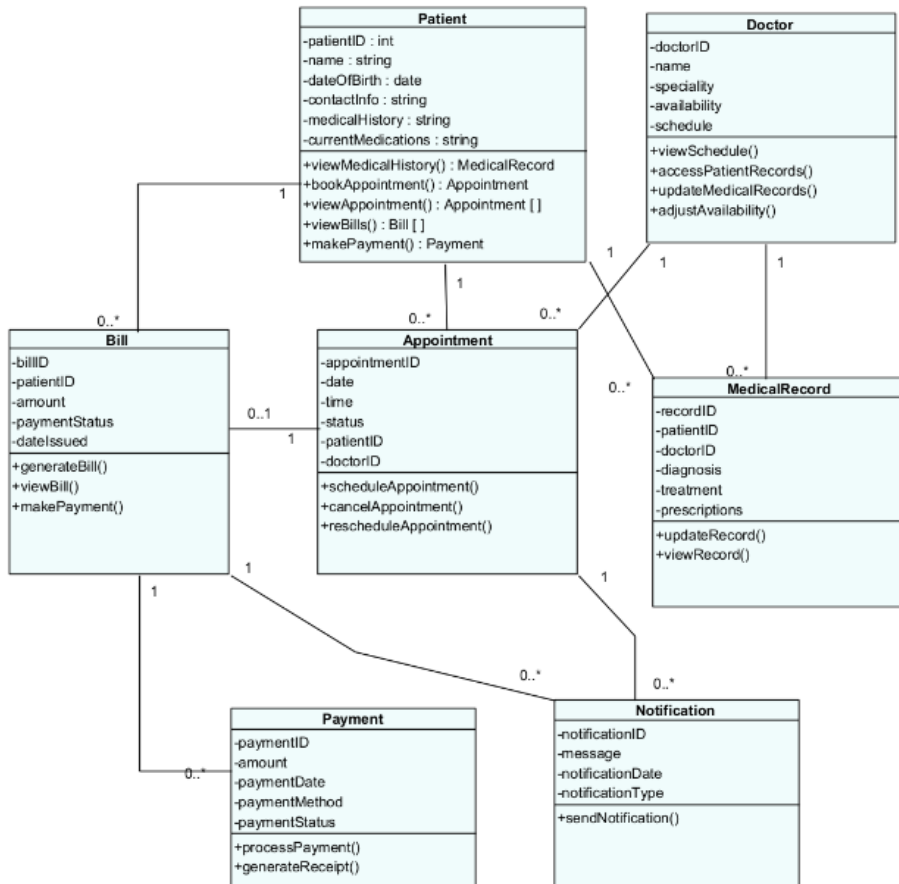
FreshGrocers needs a Sales/POS Module to process product purchases and payments directly at its stores.

2. Scheduling and Delivery Management – CleanGreen

CleanGreen needs a Scheduling and Delivery Management Module to arrange laundry pickups, deliveries, and routes. This is particularly important because its service involves collecting and returning customers' clothes.

Overall: Although both businesses need ERP to integrate their operations and manage information, their specific requirements differ because FreshGrocers sells physical products, while CleanGreen provides a service involving order processing, cleaning, pickup, and delivery.

HMS Class diagram



HealthFirst General Hospital - Updating Medical Records Sequence Diagram

